AN AGENTIC MULTIMODAL AI FRAMEWORK FOR AIRCRAFT TECHLOG INTELLIGENCE

AIMLCZG628T: Dissertation / Project Work

by

Ankit Bhardwaj

2024AA05939

Dissertation work carried out at

The Emirates Group, Dubai, UAE

Submitted in partial fulfilment of

Master of Technology (Artificial Intelligence & Machine Learning)

degree programme

Under the Supervision of

Alla Venkatesh Rao

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE

PILANI (RAJASTHAN)

June 2026

ABSTRACT

Aircraft maintenance operations at large carriers generate thousands of defect entries every month through electronic technical logbooks. Engineers working under time pressure often have to cross-reference maintenance manuals, MEL/CDL provisions, engineering orders, and previously resolved cases while moving between systems that do not talk to each other. This creates a high cognitive load, and even a delayed defect resolution can affect aircraft-on-ground time as well as operational safety.

This dissertation develops an Agentic Multimodal AI Framework, referred to throughout this report as the Techlog Copilot, to support maintenance engineers during defect investigation. The framework triages defects across eight fault categories, retrieves cited evidence from a curated knowledge corpus, predicts recurring defects on a per-tail basis, and generates explainable recommendations within safety governance constraints aligned to EASA Level 1/2 guidance.

At the mid-semester stage, I have completed the foundational data acquisition and preparation pipeline. This includes downloading and normalising four real public aviation datasets, generating a schema-consistent synthetic corpus for model training, and constructing train/validation/test splits for both the triage and recurrence tasks. The next phase will focus on implementing the RAG pipeline, training the triage and recurrence models, and integrating the agent orchestration layer.

Signature of the Student: Ankit *Date:* 16/06/2026

Signature of the Supervisor: Alla RAO *Date:* 16/06/2026

CONTENTS

1. INTRODUCTION AND PROBLEM CONTEXT

Every airline has to have a logbook which is used to record defects idenfied during the fligh or during service of the aircraft. Every defect reported by a pilot, every maintenance action completed by an engineer, and every deferred item under a MEL provision is recorded in the logbook. This data can grow into thousands of entries each week for a large international ariliner. However, this data currently serves only as a compliance and record-keeping requirement; it is not being used for predictive maintenance, anomaly pattern detection, or real-time decision support. This represents both a limitation and an opportunity.

The practical problem I set out to address is simple to state but difficult to implement: how can an engineer ask a maintenance question in natural language, receive a relevant evidence-backed recommendation.

Existing maintenance support tools at most carriers are largely retrieval-based. Engineers search PDF manuals by keyword, look up MEL items by ATA chapter, and rely on their memory for recurring fault patterns. These sources are not usually integrated, and they do not provide a predictive view of repeat defects. The research opportunity is therefore to combine retrieval-augmented generation, multimodal deep learning, and agentic AI frameworks in a way so that the constraints of an airline MRO environment are kept intact.

The proposed framework consists of five interacting components: a retrieval agent drawing on a knowledge corpus of approximately 1,213 documents, a triage agent classifying defects across eight ATA-aligned fault categories, a recurrence prediction agent operating on per-tail defect histories, a reasoning agent using a parameter-efficient fine-tuned LLM, and a safety layer for confidence gating, abstention, and audit logging. The safety governance approach is aligned to EASA's Concept Paper on Level 1 and Level 2 Machine Learning applications (Issue 2, 2023).

The dissertation project is intended as a research prototype for academic purposes. No proprietary airline data is used at any stage; all real data comes from publicly accessible sources, and the synthetic component is clearly separated throughout.

2. SYSTEM ARCHITECTURE

The framework is built as a multi-agent system orchestrated through LangGraph, with each agent responsible for a distinct function in the maintenance decision pipeline. The overall flow is shown below.

```
Engineer Query
     |
     v
Agent Orchestrator (LangGraph)
     |
     +---> Triage Agent         ---> 8-class defect label + confidence
     |     (PyTorch MLP)
     |
     +---> Recurrence Agent     ---> repeat-risk within 30 days
     |     (PyTorch LSTM)
     |
     +---> Retrieval Agent      ---> top-k cited documents from corpus
           (LangChain + Chroma)
                |
                v
          Reasoning Agent (Mistral-7B + LoRA)
                |
                v
          Safety Layer
          (confidence gate, abstention, audit log)
                |
                v
          Engineer UI (Streamlit + FastAPI)
```

Triage agent. The triage agent is a multimodal MLP with three input streams: the defect narrative text embedded through a sentence-transformer, structured metadata such as ATA chapter and flight phase, and 42 C-MAPSS-style sensor features based on last value and slope per sensor. The eight output classes map to the fault taxonomy derived from ATA iSpec 2200 chapter groupings.

Recurrence agent. The recurrence agent uses an LSTM encoder over a chronologically ordered sequence of prior defect events for each tail number. Its input features include defect category, ATA chapter, severity, and days elapsed since the last event. The binary output estimates whether the same defect category is likely to recur within a 30-day horizon.

Retrieval agent. The retrieval agent is implemented as a RAG pipeline using Chroma as the vector store and sentence-transformers for document embedding. At this stage, the knowledge corpus contains 1,213 documents spanning real FAA Airworthiness Directives, synthetic AMM/MEL/EO/ADV procedures, and de-identified case narratives from NASA ASRS.

Reasoning agent. The reasoning agent uses Mistral-7B with LoRA adapters fine-tuned on the 26,210 real defect-to-action pairs extracted from FAA Service Difficulty Reports. Given the retrieval context and triage output, it generates a ranked recommendation with citations.

Safety layer. The safety layer applies confidence gating, suppresses low-confidence outputs, and routes uncertain cases to a human escalation flag. Nine safety-critical question types in the QA benchmark are marked MUST_ABSTAIN, where the expected behaviour is to decline the answer rather than guess. Every decision, including abstentions, is written to an audit log.

3. DATA ACQUISITION AND PREPARATION — WORK DONE

The first major deliverable of the dissertation is a reproducible data preparation pipeline (`src/data/`). It downloads, normalises, and generates the datasets required for the modelling tasks. I kept the pipeline deterministic (`seed=42`) so that the same inputs produce the same train/validation/test splits and synthetic records each time it is run.

3.1 Real Public Data

Four public data sources were integrated:

NASA C-MAPSS Turbofan Degradation Dataset. The four sub-datasets (FD001–FD004) from the NASA Prognostics Center of Excellence were downloaded and extracted. These files contain multivariate run-to-failure time series for turbofan engines across varying operating conditions. Their schema, made up of three operational settings and 21 sensor measurements, is used directly in this project to build the signal-modality input for the triage classifier. Separate train, test, and RUL (remaining useful life) files are available for each sub-dataset. This data is real and publicly licensed.

FAA Service Difficulty Reports (SDR), 2023–2025. Three annual CSV exports were downloaded from the FAA's public endpoint, covering approximately 197,000 individual defect reports filed by US-registry operators. Each record follows a 76-column schema with fields such as JASC/ATA code, part name, condition, stage of operation, and a free-text Discrepancy field. In many records, that field contains both the defect description and the corrective action taken. The normalisation step maps each record onto the project's eight-class defect taxonomy using ATA chapter lookup as the primary method and keyword matching as a fallback. This produced 196,118 normalised records, of which 166,037 were successfully mapped to one of the eight classes. The remaining 30,081 records fell into an OTHER bucket. I have retained these records with their mapping method flagged, since they show a clear limitation of the current heuristic approach.

Corrective action extraction from SDR free text was one of the noisier parts of the pipeline. The Discrepancy field often combines defect narrative and action text in a single block, separated by phrases such as "CORRECTIVE ACTION:", "C/A", or "RECTIFICATION". I used a regex-based splitter for these markers and extracted 26,210 defect-to-action pairs where a parseable corrective action was present. These pairs provide the primary supervised signal for fine-tuning the reasoning agent.

NASA ASRS Report Sets. Three curated report-set PDFs were downloaded from NASA's Aviation Safety Reporting System, covering maintenance technician reports, fuel management issues, and cabin fumes/smoke incidents. Each PDF contains 50 de-identified safety reports submitted voluntarily by aviation personnel. The PDFs were parsed using `pypdf`, with individual report blocks segmented by ACN (Accession Number) markers. Metadata fields such as aircraft type, component, flight phase, and anomaly type were extracted from the structured header of each report, while the free-text Narrative section was parsed separately. This produced 150 structured records that were added to the knowledge corpus as CASE-type documents.

FAA Airworthiness Directives. 500 AD rules were retrieved via the Federal Register API, with full text for 49 of these fetched from govinfo.gov. The remaining 451 carry only the abstract text, as the Federal Register's raw text endpoint is behind access controls. All 500 AD documents are included in the RAG corpus.

3.2 Synthetic Data

A schema-consistent synthetic corpus was generated to complement the real public data. I treated the synthetic component as supporting material, not as a replacement for real data. Its main role is to provide labelled training material for modelling tasks where public aviation data does not exist in the required form.

Synthetic techlogs. 4,425 electronic techlog records were generated across 70 tail numbers styled on an Emirates-style A6- registry fleet (A320, A321, A330, A350, B737, B777, B787 types). Each record includes the defect category, ATA chapter, component, severity, flight phase, station, and two versions of the defect narrative: a raw form with shorthand abbreviations typical of real techlog language, for example "Gen CB tripped repeatedly dur approach PLS ADV", and a clean normalised form. Per-tail histories are internally consistent, with recurrence flags assigned using 30-day lookback logic across the chronological sequence.

Knowledge corpus. 563 synthetic documents were generated across six document types: 42 AMM procedure snippets, 16 MEL deferral entries, 40 AD documents, 35 Engineering Orders, 30 operational advisories, and 400 prior-case documents. Each document references a specific defect category and ATA chapter, and the component-to-ATA-chapter pairing was validated programmatically (zero mismatches). The 500 real FAA ADs and 150 ASRS reports were appended to this base corpus, bringing the total to 1,213 retrievable documents.

C-MAPSS-style signal windows. 3,915 signal windows were generated following the NASA C-MAPSS schema (3 operational settings + 21 sensor measurements). Each window is associated with a techlog record and has a defect-category-specific degradation signature injected: pressure drift for hydraulic and landing gear faults, flow anomaly for fuel system faults, temperature rise for bleed air faults, and similar patterns for the remaining categories. A sensor summary file provides fusion-ready per-sensor last-value and slope features for each window.

QA benchmark. 189 question-answer pairs covering all eight defect categories were generated with a safety-aware taxonomy: 163 MUST_CITE items (answer requires citing a specific corpus document), 17 CAN_ANSWER items (answerable from general knowledge), and 9 MUST_ABSTAIN items (safety-critical authority questions where the correct response is to decline and escalate). This benchmark will be used to evaluate retrieval Recall@k, nDCG, citation correctness, hallucination rate, and abstention precision/recall.

3.3 Train/Val/Test Splits

Splits were generated chronologically to prevent temporal leakage. This is especially important for the recurrence task, where using future events to predict past ones would give artificially inflated metrics. The train set covers the earliest 80% of dates, the validation set the next 10%, and the test set the final 10%.

For triage: 3,540 train / 442 validation / 443 test records. Class balance in the training set is roughly uniform across the eight categories (400–480 per class). 3,888 of the total examples carry the signal modality.

For recurrence: 3,470 train / 372 validation / 373 test records. Positive rate (recurrence within 30 days) is approximately 37% in the training set.

4. DATASET DESCRIPTION AND STATISTICS

Dataset	Type	Path	Volume
NASA C-MAPSS FD001–FD004	Real	`data/raw/nasa_cmapss/`	14 files (train/test/RUL per sub-dataset)
FAA SDR 2023–2025	Real	`data/raw/faa_sdr/real/`	~197,000 rows, 76 columns
NASA ASRS (3 report sets)	Real (de-identified)	`data/raw/nasa_asrs/report_sets/`	3 PDFs, 150 parsed reports
FAA Airworthiness Directives	Real	`data/raw/faa_ad/real/`	500 metadata records, 49 full-text
Synthetic Techlogs	Synthetic	`data/raw/techlogs/`	4,425 records, 70 tail histories
Knowledge Corpus	Mixed	`data/raw/knowledge_corpus/`	1,213 documents total
C-MAPSS-style Signal Windows	Synthetic	`data/raw/cmapss_signals/`	3,915 windows, 117,450 cycle-rows
Triage Splits	Processed	`data/processed/triage/`	3,540 / 442 / 443
Recurrence Splits	Processed	`data/processed/recurrence/`	3,470 / 372 / 373
SDR Normalised Records	Processed	`data/processed/`	196,118 records
Defect-Action Pairs	Processed	`data/processed/`	26,210 pairs
QA Benchmark	Processed	`data/processed/`	189 items

Table 1: Dataset Summary

A few observations from working with this data are worth noting. The FAA SDR free text is genuinely noisy: it comes in upper case, uses heavy abbreviations, and often runs the defect and corrective action together without a clean separator. The regex-based splitter handles the majority of cases, but approximately 10% of records with corrective actions were not cleanly separable and were excluded from the pairs file. Given the overall volume, this is an acceptable loss for the current stage.

The ASRS PDF structure is generally well-suited to automated extraction. Each report follows a consistent tabular layout with explicit field labels, and the Narrative section is clearly delimited. The main challenge was that `pypdf` occasionally merged adjacent text blocks when the original PDF layout was column-based. A few reports therefore had malformed metadata fields, although the narrative text was consistently recoverable.

The 30,081 records that fell into the OTHER taxonomy bucket represent a real limitation of the heuristic ATA mapping approach. Many of these cover ATA chapters not in the project's 15-chapter scope (e.g., ATA 25 cabin equipment, ATA 45 central maintenance system). These records are preserved in the normalised output with the mapping_method field flagged as "unmapped", so future work could apply an ML-based classifier to recover them.

5. PROGRESS REPORT

Sl No	Phase	Start Date – End Date	Work to be Done	Status
1	Dissertation Outline	05 May 2026 – 10 May 2026	Literature review and prepare Dissertation Outline	COMPLETED
2	Data Acquisition & Preparation	10 May 2026 – 15 Jun 2026	Download public datasets; generate synthetic corpus; build train/val/test splits	COMPLETED
3	RAG Pipeline & Vector Store	16 Jun 2026 – 05 Jul 2026	Embed knowledge corpus into Chroma; implement retrieval agent; evaluate Recall@k and nDCG on QA benchmark	IN PROGRESS
4	Triage & Recurrence Models	06 Jul 2026 – 20 Jul 2026	Train multimodal MLP triage classifier (text + metadata + signals); train LSTM recurrence model; ablation baseline comparison	PENDING
5	LLM Fine-tuning & Agent Integration	21 Jul 2026 – 06 Aug 2026	LoRA fine-tune Mistral-7B on defect-action pairs; integrate triage, recurrence, retrieval, and reasoning agents in LangGraph; implement safety layer with confidence gating and abstention	PENDING
6	API, UI & Evaluation	07 Aug 2026 – 13 Aug 2026	FastAPI backend; Streamlit engineer UI; full system evaluation against all metrics (AUROC, nDCG, hallucination rate, latency p95)	PENDING
7	Dissertation Review & Submission	14 Aug 2026 – 19 Aug 2026	Submit dissertation to Supervisor and Additional Examiner for review; final corrections and submission	PENDING

6. ABBREVIATIONS USED

Abbreviation	Full Form
AD	Airworthiness Directive
AHM	Aircraft Health Monitoring
AMM	Aircraft Maintenance Manual
ASRS	Aviation Safety Reporting System (NASA)
ATA	Air Transport Association (chapter numbering standard)
AUPRC	Area Under Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BITE	Built-In Test Equipment
CDL	Configuration Deviation List
C-MAPSS	Commercial Modular Aero-Propulsion System Simulation
CMC	Central Maintenance Computer
EASA	European Union Aviation Safety Agency
EGT	Exhaust Gas Temperature
EO	Engineering Order
FAA	Federal Aviation Administration
GPU	Graphics Processing Unit
JASC	Joint Aircraft System/Component
LLM	Large Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory (recurrent neural network variant)
MEL	Minimum Equipment List
MLP	Multilayer Perceptron
MRO	Maintenance, Repair and Overhaul
nDCG	Normalised Discounted Cumulative Gain
NLP	Natural Language Processing
QA	Question Answering
RAG	Retrieval-Augmented Generation
RUL	Remaining Useful Life
SDR	Service Difficulty Report
SRM	Structural Repair Manual

7. REFERENCES

[1] Lewis, P., Perez, E., Piktus, A., et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *NeurIPS*, 2020.

[2] Vaswani, A., Shazeer, N., Parmar, N., et al. "Attention Is All You Need." *NeurIPS*, 2017.

[3] Hu, E. J., Shen, Y., Wallis, P., et al. "LoRA: Low-Rank Adaptation of Large Language Models." *ICLR*, 2022.

[4] EASA Concept Paper: First Usable Guidance for Level 1 & 2 Machine Learning Applications, Issue 2, European Union Aviation Safety Agency, 2023.

[5] NIST AI Risk Management Framework (AI RMF 1.0), National Institute of Standards and Technology, 2023.